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A decentralized credit risk prediction method based on federated cost-sensitive learning☆

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ABSTRACT

With the expansion of business in the financial sector and the improvement of storage technologies, the industry has generated a large amount of credit-risk-related data, which is expected to further enhance credit risk prediction. However, the direct centralization of all relevant data to train credit risk prediction models in the industry carries the risk of privacy breaches. We develop a secure federated cost-sensitive learning method with a cloud model (FedCSL-CM) for joint risk control among financial organizations. We perform comparative experiments on four credit datasets, assess the security and effectiveness of the proposed method in leveraging multi-source data and improving the performance of the decentralized credit risk prediction method, and demonstrate that the proposed method can facilitate data sharing and decentralized modeling among different financial organizations to achieve more efficient credit risk management.

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1. Introduction

In recent years, the prevention of systemic financial risks has been a significant concern for society and the financial sector. Credit evaluation is an effective approach to helping financial organizations manage credit risk, which centers on the construction of powerful credit risk prediction models to identify potential default risks of borrowers (Chen et al., 2025; Hilscher & Wilson, 2017). With the booming development of artificial intelligence techniques, machine learning algorithms are increasingly being applied to the development of credit risk prediction models (Bellotti & Crook, 2013; Xiao et al., 2023). Machine learning algorithm-based credit risk prediction models can extract useful information from a large amount of historical credit data (including personal information, financial status, and payment history) to identify potential default risks of borrowers to distinguish the creditworthiness of borrowers accurately, thereby improving the efficiency of credit decision-making.

To obtain a robust credit risk prediction model, the training process often requires feature mapping and parameter optimization on large amounts of historical transaction data. However, most financial organizations possess only a small amount of credit data, which prevents credit risk prediction models from performing as expected. With society's concern for data privacy and safety, countries have issued appropriate data protection regulations that impose severe restrictions on the use of data (He et al., 2023). These render it impractical for

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financial organizations to share and exchange data freely with other organizations, resulting in the formation of data silos, thus hindering the further development of credit risk prediction. Therefore, developing an approach that enables decentralized credit risk data modeling while safeguarding privacy is of significant importance.

The emergence of federated learning has given a novel impetus to the development of decentralized credit risk prediction. Federated learning (McMahan et al., 2017) is a secure distributed machine learning framework that allows multiple parties to keep their data locally while exchanging only model parameters, thereby significantly reducing the risk of data leakage. For the financial industry, financial organizations in different regions have different customer groups but similar business patterns, so the data characteristics they collect are generally identical or highly similar. Therefore, horizontal federated learning (HFL)—a federated learning framework designed for horizontally partitioned data scenarios—offers potential advantages for data sharing and decentralized modeling among different financial organizations. Unbalanced data distribution is also a primary factor affecting the discriminative performance of credit risk prediction models. This typically means that the number of non-default (good credit) applicants is much larger than that of default (bad credit) applicants (Crone & Finlay, 2012). In decentralized settings, data imbalance remains a persistent issue, and discrepancies in data distributions across different financial organizations further exacerbate this challenge. In traditional credit risk prediction tasks, the dominant methods for solving the class imbalance issue are resampling methods, cost-sensitive learning methods, and hybrid methods. Although class imbalance in credit risk prediction has been addressed well by existing studies, class imbalance in decentralized credit risk prediction, which requires privacy protection, has not yet received attention. After careful analysis, we conclude that decentralized credit risk prediction currently suffers from the following challenges:

(1) Few studies are focusing on the field of decentralized credit risk prediction, and the training process of traditional centralized credit risk prediction methods is performed on a single node, without considering data sharing among multiple clients to co-train the credit risk prediction model. As a result, existing credit risk prediction methods cannot provide a valuable reference for decentralized credit risk prediction. The field is still in its infancy.
(2) Mainstream unbalanced learning methods mostly require analyzing the distribution of training data before they can be performed (Tian et al., 2024). In decentralized credit risk prediction, clients cannot directly upload any information related to their private data because of the strict requirements for data privacy protection. Consequently, these unbalanced learning methods are limited by the privacy requirements of decentralized credit risk prediction and cannot be applied well in this domain. This makes the class imbalance issue more prominent for decentralized credit risk prediction scenarios that require privacy protection.
(3) In federated learning scenarios, large parameter drifts may occur in local models trained on different clients, especially when class imbalance leads to heterogeneous data distribution. The classical horizontal federated learning approach averages the local model parameters of participants to obtain the global model, which leads to an overall poor training result.
(4) Federated learning utilizes gradients and model parameters to train shared models collaboratively, and thereby ensures that the data privacy of clients is not compromised. In practical applications, malicious attackers may still infer sensitive information by backpropagating intermediate data (i.e., model parameters) through membership inference attacks (MIA; Shokri et al., 2016) or other techniques, potentially leading to the leakage of clients' private data.

Therefore, we propose a secure federated cost-sensitive learning method with a cloud model (FedCSL-CM) to address the aforementioned problem. Specifically, to utilize multi-source data samples while protecting data privacy, we propose a safe decentralized credit risk prediction method based on the HFL framework to enhance the performance of credit risk prediction. First, to solve the imbalance between the distribution of positive and negative samples, we develop a local model parameter update method based on cost-sensitive learning (CSL) to enhance the classification performance of the decentralized credit risk prediction method on minority samples. The Cloud Model (CM) is a transformation model that handles qualitative concepts and quantitative representations, offering potential advantages for evaluating the performance of different local models. Then, to address the issue of local model parameter drift caused by class imbalance, we further develop a CM-based adaptive local model weighting strategy to assign appropriate weights to local models, guiding the global model's update direction toward the optimal solution. Finally, to prevent data privacy from being stolen by malicious attackers, we construct a secure model aggregation method based on a homomorphic encryption (HE) algorithm to achieve secure global aggregation, thereby further enhancing the security of the decentralized credit risk prediction method.

The main contributions of this work are summarized as follows: We present a secure and decentralized credit risk prediction method, FedCSL-CM, that effectively models and protects unbalanced credit data under a horizontal federated learning framework. Specifically, we embed a cost-sensitive learning function proposed in this paper into the framework to enable the model to better adapt to unbalanced data distributions. Meanwhile, we design a CM-based adaptive local model weighting strategy to enhance the robustness and forecast accuracy of the global model. Furthermore, to ensure data privacy and security, we integrate a homomorphic encryption technique to support secure model aggregation. Finally, extensive experimental results demonstrate that the proposed method outperforms existing benchmark

federated learning methods in both performance and security. Methodologically, this study advances decentralized learning in financial risk management by addressing the dual challenge of privacy and data imbalance. From a policy perspective, the framework provides a viable path for regulators and financial organizations to implement privacy-compliant, cooperative credit risk prediction systems under data protection laws. These insights can guide future regulatory sandbox initiatives and collaborative risk assessment mechanisms.

The structure of this study is as follows. Section 2 gives a review of relevant literature. Section 3 presents the proposed FedCSL-CM method. Section 4 provides the experimental design. Section 5 presents the analysis and discussion of experimental results. The final section discusses the conclusions and directions for future work.

2. Literature review

2.1. Federated learning

Federated learning, a framework of decentralized machine learning algorithms that protects data privacy, has attracted considerable attention from academia and industry and has been applied to a variety of domains (Bercea et al., 2022; Feng et al., 2024; Tao et al., 2023; Wu et al., 2022c). According to different data-sharing perspectives, federated learning is commonly classified into three types: HFL (sharing samples), vertical federated learning (VFL; sharing features), and federated transfer learning (FTL; sharing samples and features) (Yang et al., 2019). Given the application potential, technical development complexity, and deployment cost, most scholars have focused their efforts on HFL and VFL.

2.1.1. Horizontal federated learning

HFL is a federated learning framework developed for mobile device environments, which can utilize data with the same features from different devices to jointly train models (McMahan et al., 2017). The federated averaging (FedAvg) algorithm is one of the best-known algorithms in horizontal federated learning, which trains a shared model by calculating the average of the local model parameters shared by each participant. In recent years, most of the research on horizontal federated learning has been developed based on the FedAvg algorithm. For example, Li et al. (2020) modified the parameters of the horizontal federated averaging algorithm FedAvg and proposed the FedProx method to address heterogeneity in federated networks. With the theoretical analysis and empirical evaluation, the FedProx method can significantly enhance the robustness of HFL in heterogeneous settings. Likewise, based on the FedAvg algorithm, Karimireddy et al. (2020) introduced a control variable in the server and clients, respectively, and proposed a new federated algorithm, SCAFFOLD, that mitigates client drift by correcting the local model update. To improve the fairness and accuracy of HFL, Huang et al. (2022) designed a new federated algorithm, FedFa, which combines dual-momentum gradient optimization and fair weight selection schemes. Yue et al. (2023) developed a fair federated learning framework, GIFAIR-FL, that adaptively adapts client weights to promote group and individual fairness and achieve good prediction accuracy. A promising algorithm, FedLGA, was proposed by Li et al. (2023), which solves the across-device heterogeneity issue by bridging the divergence of local model updates through gradient approximation. Yan et al. (2023) developed FedLaAvg, a federated algorithm that takes into account intermittent client availability, and the theoretical analysis and experimental results indicated a more stable performance of their approach compared to FedAvg.

Regarding financial applications, Yang et al. (2022) developed a privacy-preserving credit data and model sharing architecture by integrating HFL and blockchain, enabling secure and efficient credit modeling and verification through a deletable Bloom filter and smart contracts. Yang et al. (2024) designed an HFL-based interpretable credit risk prediction system and empirically analyzed the effectiveness, security, and interpretability of their proposed system.

2.1.2. Vertical federated learning

VFL is another federated learning framework that enables participants to collaboratively train models using different sample features (Hardy et al., 2017). Currently, feature segmentation learning frameworks based on tree models (Cheng et al., 2021), linear regression models (Gratton et al., 2018; Xu et al., 2021; Yang et al., 2019), and neural network models (Hu et al., 2019) have been developed. For example, for the task of VFL of feature distributions, Hu et al. (2019) proposed an asynchronous stochastic gradient descent method and demonstrated its effectiveness and theoretical convergence in logistic regression and neural networks. Zhang et al. (2022) explored the impact of unbalanced features on VFL and proposed a new learning approach to enhance the VFL performance on unbalanced features by incorporating the number of features possessed by each participant and their computational efficiency. In addition, they designed an optimization issue to reduce the training period of VFL, which improves the training efficiency by adaptively choosing the number of local updates. To address the label scarcity problem, Shen et al. (2024) proposed an unsupervised vertical federated learning (FedLED) approach to enhance the model's learning effect by utilizing an unsupervised model for knowledge transfer learning on the unlabeled target domain.

Regarding financial applications, He et al. (2023) formulated a decentralized logistic regression model via the VFL framework and demonstrated experimentally that their proposed model can exploit multiple source information efficiently for credit risk prediction. In our published study, Wang et al. (2025) proposed a vertical logistic regression method with cost-sensitive learning and analyzed the interpretability, privacy, complexity, and communication cost of this method.

Existing research has shown that federated learning techniques can effectively address data security issues in the decentralized credit risk prediction modeling process. However, these studies have not examined the impact of unbalanced data distributions on the performance of federated learning-based decentralized credit risk prediction

models. Furthermore, although federated learning safeguards participants' data privacy through gradient and model parameter exchange, potential risks of sensitive information theft via malicious attacks persist in practical applications.

2.2. Imbalanced learning

Class imbalance has been a predominant problem in the domain of credit risk prediction. This label-skewed distribution poses a great barrier to the training of credit risk prediction models, thus resulting in poor performance (Liu et al., 2008). Therefore, researchers have conducted many works on unbalanced credit risk prediction in recent years, which are mainly categorized into three solutions: data-level methods, algorithm-level methods, and hybrid methods.

2.2.1. Resampling methods

Resampling methods generally change the class distribution of the original dataset through resampling strategies, mainly including undersampling and oversampling. Random undersampling is one of the most basic undersampling methods and is implemented by randomly deleting a certain percentage of the original majority samples to achieve a balance between the two classes (Kovacs, 2019). For example, Ng et al. (2015) used a clustering algorithm to analyze the distribution information of the majority samples and filtered the samples from the majority samples through a stochastic sensitivity measure, thereby ensuring a balanced dataset. Niu et al. (2020) employed an undersampling method to balance credit datasets based on the distribution of majority samples, and the experimental results on several datasets demonstrated that their method has robust predictive performance. Abedin et al. (2023) introduced a WSMOTE-ensemble approach for small business credit risk modeling, which significantly improves default prediction performance under unbalanced data conditions.

Jiang et al. (2023) considered the GAN as a sample generation tool to deal with the class imbalance problem and experimentally showed its potential application in solving unbalanced credit risk prediction. Xiao et al. (2024) developed a deep integration method, VAE-DF, based on a variational autoencoder and a deep forest for improving the performance of credit risk prediction. Chai et al. (2025) developed a feature-dimension–based credit risk evaluation model for farmers, integrating OPSO-GINI feature selection and the META-SAMPLER method to improve loan default identification.

2.2.2. Cost-sensitive learning methods

Cost-sensitive learning methods reduce the bias of the classification model toward majority class samples at the algorithm design level by reconsidering the asymmetric misclassification cost between majority and minority class samples (Zhu et al., 2020). For unbalanced credit risk prediction, the common practice is to design a loss function with the optimization goal of reducing the total misclassification costs, so that the classification algorithm focuses on the minority class samples with a high misclassification cost. For example, Maldonado et al. (2017) presented a profit-driven support vector machine for unbalanced credit risk prediction by considering both variable acquisition costs and misclassification costs and accurately identifying profits, which was practiced on a real dataset from the Chilean bank and achieved excellent performance. Xiao et al. (2020) developed a cost-sensitive neural network model based on semi-supervised learning, and experimental results proved their approach's superior performance on unbalanced credit datasets. Liu et al. (2022) considered a focal loss function and improved the tree model to solve the class imbalance issue in credit risk prediction. Petrides et al. (2022) evaluated the effectiveness of different cost-sensitive loss functions in credit risk prediction and corroborated their proposal in a real case study. Khalili and Rastegar (2023) tested the performance of the defined cost-sensitive objective function on five classifiers with several evaluation metrics, and the experimental results showed the effectiveness of their proposed model in improving the performance of unbalanced credit risk prediction. Yuan et al. (2025) introduced a weighted average ensemble model that minimizes misjudgment costs and employed a two-stage feature selection process to prevent overfitting, thereby enhancing financial distress prediction performance while reducing data acquisition costs. Xia et al. (2025) proposed a two-stage credit scoring model that integrates large language models and an improved FocalPoly loss function to jointly address data scarcity and class imbalance in fintech lending, achieving superior predictive performance and interpretability.

2.2.3. Hybrid methods

Hybrid methods are effective combinations of several different imbalance processing approaches to obtain better prediction results. The dominant strategy is to integrate resampling techniques with ensemble learning methods (Zhao et al., 2023). Kabir et al. (2018) embedded the classification difficulty of training samples into a bagging strategy that varies the sampling probability of each sample according to the classification difficulty, thus improving the model's classification performance on unbalanced datasets. Seng et al. (2021) proposed an unbalanced learning method, NUS-SE, based on the domain undersampling technique and stacking ensemble strategy, which can select the majority class samples based on the domain information and embed the undersampling in the stacking ensemble, thus ensuring that the classification model's sensitivity can be improved without decreasing the prediction accuracy. Zhao et al. (2023) developed a new resampling method, BCWF, to deal with the class imbalance problem from the perspective of the sample level. Park and Cheng (2024) proposed a hybrid resampling method based on cluster impurity, and they considered both intra-cluster class imbalance and inter-cluster overlap problems, thus improving the imbalance learning performance. To address the issue of class imbalance in consumer credit data, Weng et al. (2025) selected appropriate sampling methods based on data characteristics and developed a Bayesian model averaging framework integrated with SHAP-based interpretability techniques.

This approach enhances both the robustness and interpretability of default prediction and further validates the value of soft information in credit risk forecasting.

Although the above methods work well to minimize the adverse impact of class imbalance, none of them consider how to leverage multi-source unbalanced data safely to jointly train credit risk prediction models. Furthermore, in privacy-preserving scenarios, the clients are unaware of each other's data distributions. As a result, some mainstream unbalanced learning methods cannot perform effectively. This makes realizing imbalance learning while preserving privacy an important challenge in the development of decentralized credit risk prediction.

3. The proposed FedCSL-CM method

3.1. Basic framework

An increase in the volume of training data is crucial for improving credit risk prediction performance (Zhou et al., 2021). However, directly incorporating data from multiple sources into the training process of a credit scoring model may lead to potential privacy leakage. Consequently, centralized credit risk modeling approaches proposed in previous works are inadequate for addressing the challenges considered in this paper. To overcome these limitations, this paper proposes a secure FedCSL-CM method, which enables collaborative modeling on distributed and unbalanced credit datasets, thereby enhancing both the accuracy and privacy protection of credit risk prediction.

The primary objective of credit risk prediction is to effectively differentiate the creditworthiness of borrowers to control the risk of default within a reasonable range (Shen et al., 2022). Therefore, it is usually formulated as a binary classification problem. To describe the process of decentralized credit risk prediction, we consider a decentralized modeling scenario with K clients and a central server. Specifically, we suppose that the client k ($k = 1, 2,\ldots, K$) has a local dataset $D_k = (x^i, y^i)|i = 1, 2, \ldots, n$, n represents the number of samples in the local dataset, x^i represents the ith sample in the local dataset, and $y^i \in (0, 1)$ is the true label corresponding to the sample x^i. In this study, we focus on the training process of HFL, so we assume that the local sample features held by each client are aligned.

As illustrated in Fig. 1, the proposed FedCSL-CM approach involves three primary steps: CSL-based local model parameter update, CM-based adaptive local model weighting, and HE-based secure model aggregation.

(1) CSL-based local model parameter update: Each client utilizes the proposed cost-sensitive learning function to perform the local model parameter updates and uploads the updated local model parameters to the server, which contributes to improving the model's learning performance on minority samples.
(2) CM-based adaptive local model weighting: the server utilizes the cloud model to evaluate the comprehensive learning performance of local models adaptively and assign their weights in the global aggregation, which helps to alleviate the problem of local model parameter drift formed by class imbalance.
(3) HE-based secure model aggregation: The server aggregates the local model parameters securely using homomorphic encryption, generating a new round of the global model and further enhancing the security of FedCSL-CM.

We explain the above steps thoroughly in Sections 3.2, 3.3 and 3.4

3.2. CSL-based local model parameter update

In credit risk prediction tasks, there are often far more non-default (good credit) examples than default (bad credit) examples, which causes the classification model to fail to identify loan applicants with default risk effectively. The unbalanced data distribution makes it more difficult to train the decentralized credit risk prediction model. The traditional cross-entropy (CE) loss function does not discriminate between the weights of positive and negative samples during the training process and thus fails to achieve good learning results for positive samples, as expressed in the equation

$$L_{CE} = -\sum_{i=n}^{n}[y^i log(p^i) + (1 - y^i) log(1 - p^i)], \quad (1)$$

where y^i is the true label of the ith sample, and p^i is the predicted probability of the ith sample.

Moreover, the local model update of the traditional federated algorithm starts by copying the global model parameters, which may also fail to guarantee the overall training quality. Therefore, as described previously, we embed a cost-sensitive learning function based on focal loss and knowledge distillation into the horizontal federated framework to enhance the learning performance of local models when dealing with unbalanced data distributions. Focal loss was initially developed by Lin et al. (2017) to address the imbalance of positive and negative samples in classification tasks. Focal loss is implemented by modifying the CE loss function, specifically by introducing an additional tunable parameter, a focusing parameter, to reduce the loss of simple samples and enable the model to concentrate more on learning from difficult samples, as expressed in the equation

$$L_f = -\sum_{i=n}^{n}[y^i(1-p^i)^\gamma log(p^i)+(1-y^i)(p^i)^\gamma log(1-p^i)], \quad (2)$$

where $(1 - p^i)^\gamma$ is a modulating factor with focusing parameter γ to balance the importance of easy and hard examples.

In addition, to further optimize the update of local model parameters, we add a new weighting factor to control the training weights of the majority and minority class samples. The focal loss function for client k can be represented as

$$L_f = -\sum_{i=n}^{n}[\alpha y^i(1-p^i)^\gamma log(p^i)+(1-\alpha)(1-y^i)(p^i)^\gamma log(1-p^i)], \quad (3)$$

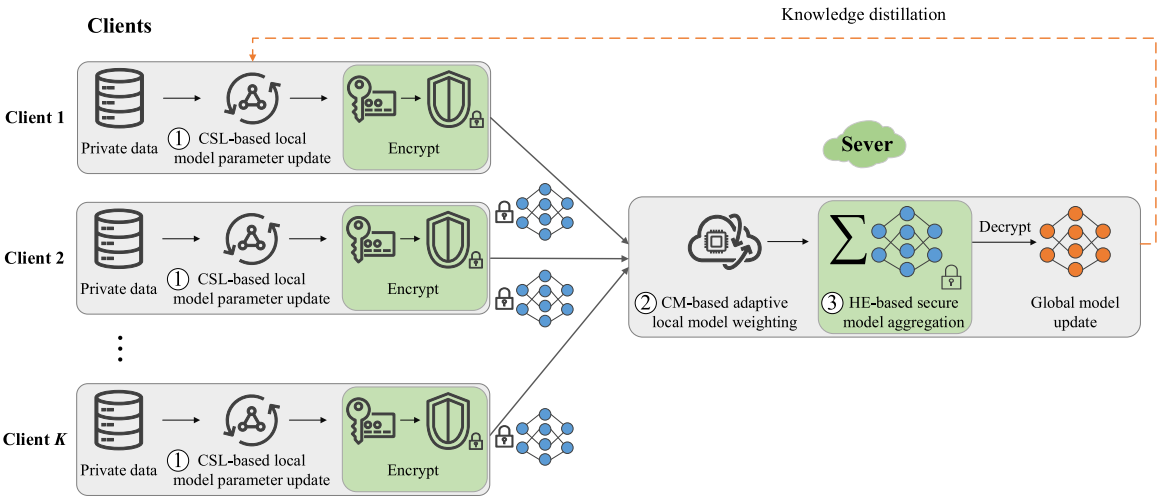


Fig. 1. Framework of the proposed FedCSL-CM method.

where $\alpha \in [0, 1]$ is a weighting factor to modulate the weights of positive and negative samples.

Knowledge distillation, proposed by Hinton et al. (2015), is an effective model compression technique that can transfer knowledge from the teacher model (a fully trained and well-performing model) to the student model (a to-be-trained or simpler model), thereby improving the performance of the student model to a level approximating that of the teacher model. Specifically, the knowledge distillation technique is implemented by adding a distillation loss term, L_{soft}, to force the student model to learn the probability outputs of the teacher model, where L_{soft} typically selects the KL divergence or mean squared error (MSE) loss to measure the distributional difference of model outputs (Wu et al., 2022b). In our proposed decentralized credit risk prediction method, the global model encapsulates information from all local models and is thus employed as a teacher model to guide local model training. Ultimately, the local loss function and local update process for client k can be expressed as

$$L_k = \lambda L_{soft}(q_{gk}, q_{lk}) + (1 - \lambda)L_f, \tag{4}$$

$$q_{gk} = \frac{exp(z^1_{gk}/\tau)}{exp(z^0_{gk}/\tau) + exp(z^1_{gk}/\tau)}, q_{lk} = \frac{exp(z^1_{lk}/\tau)}{exp(z^0_{lk}/\tau) + exp(z^1_{lk}/\tau)}, \tag{5}$$

$$w^{t+1}_k = w^t_k - \eta \nabla L_k(w^t_k, q^t_{gk}), \tag{6}$$

where L_k is the local loss function for client k, λ is a relative weight parameter, L_{soft} is a distillation term to quantify the distributive differences between models' outcomes, q_{gk} and q_{lk} are the probability outcomes of the global and local model for client k, z^0_{gk} and z^1_{gk} are the logit outputs of the global model for classes 0 and 1, z^0_{lk} and z^1_{lk} are the logit outputs of the global model for classes 0 and 1, w^{t+1}_k and w^t_k are the local model parameters for the $(t+1)$th and tth rounds of client k, η is the learning rate, and τ ($\tau \geq 1$) is a distillation temperature parameter; the higher its value the smoother the probability distribution.

3.3. CM-based adaptive local model weighting

Generally, there are significant parameter drifts between local models trained by different clients on their private datasets, especially with the unbalanced data distribution. The federated averaging algorithm directly averages the weights for aggregation, which deviates from practice and leads to poor model performance. Consequently, we develop an adaptive local model weighting strategy based on the cloud model to more appropriately distribute the weights of local models for improving the global aggregation effect. The cloud model is a mathematical model that integrates fuzzy set theory, probability theory, and mathematical statistics, which can be applied to deal with the inter-transformation between fuzzy concepts and quantitative indicators (Liu et al., 2023b). A backward cloud generator (BCG) is a classical cloud generation algorithm that can realize the transformation of concrete values to qualitative concepts as expressed by digital features with expectation *Ex*, entropy *En*, and hyper entropy *He*. Therefore, we use the BCG and assign the weight of each local model in global aggregation based on its training performance. The detailed process of adaptive local model weighting based on the BCG (considering K local models and m performance measures) is as follows:

(1) Construct a system of performance measures and a decision matrix of performance measures about local models.
(2) The decision matrix of local model k under performance measures U_j ($j = 1, 2, \ldots, m$) is transformed to a standard matrix $V = (v_{kj})_{k\times m}$, where $V^+ = v^1_{max}, v^2_{max}, \ldots, v^m_{max}$ and $V^- = v^1_{min}, v^2_{min}, \ldots, v^m_{min}$ represent positive and negative ideal models, respectively.
(3) Compute the cloud models for local model k, positive and negative ideal models $N_k(Ex, En, He)$, $N^+(Ex, En, He)$, and $N^-(Ex, En, He)$, where the digital characteristics *Ex*, *En*, and *He* can be expressed by

the following formulas:

$$\overline{V} = \frac{1}{m}\sum_{j=1}^{m} v_{kj}, \tag{7}$$

$$M_1 = \frac{1}{m}\sum_{j=1}^{m} \left| v_{kj} - \overline{V} \right|, \tag{8}$$

$$S^2 = \frac{1}{m-1}\sum_{j=1}^{m} (v_{kj} - \overline{V})^2, \tag{9}$$

$$Ex = \overline{V}, En = \sqrt{\frac{\pi}{2}} \times M_1, He = \sqrt{\left| S^2 - En^2 \right|}. \tag{10}$$

(4) Compute the similarity between the local model and the positive and negative ideal models, respectively. The similarity between any two cloud models can be calculated according to the following equations, where $\Phi(\varepsilon)$ is a standard normal distribution function:

$$S(N_k, N_b) = \frac{1}{2} + \frac{1}{2\Phi(\varepsilon)} - \Phi(\varepsilon), \tag{11}$$

$$\varepsilon = \frac{|Ex_k - Ex_b|}{\sqrt{En_k^2 + He_k^2} + \sqrt{En_b^2 + He_b^2}}. \tag{12}$$

(5) Compute the relative similarity of local model k. A higher value of the relative similarity represents a better overall performance of the model, which is formulated as

$$D(k) = \frac{S(N_k, N^+)}{S(N_k, N^+) + S(N_k, N^-)}. \tag{13}$$

(6) Determine the weight of each local model based on the relative similarity value obtained from Eq. (13), which is calculated as

$$weight_k = \frac{D(k)}{\sum_{k=1}^{K} D(k)}. \tag{14}$$

3.4. HE-based secure model aggregation

In this study, we assume that the clients and central server are "honest and curious" semi-trustworthy entities. In a federated learning scenario, the semi-honest model is a hypothetical model that is consistent with practice. With this assumption, both clients and the central server may make attempts to access other clients' private data by inferring intermediate data (i.e., model parameters) during the training period, while adhering to the cooperation agreement. Although each client's private data are kept local in the above method, the model parameters are transmitted in plaintext, which may potentially compromise client privacy. Therefore, to further protect data privacy, we employ a homomorphic encryption scheme to encrypt the local model parameters in the decentralized modeling process and perform a secure computation that allows the server to access only the decrypted global parameters. The specific method used is the Paillier algorithm (Paillier, 1999), which is an additive homomorphic encryption algorithm.

We extend the Paillier algorithm to the encrypted computation of local model parameters for secure model aggregation. For simplicity, we use $|\cdot|$ to denote the ciphertext scalar, and the specific procedure of the HE-based secure model aggregation method is as follows: first, the server generates the key pairs for the homomorphic encryption algorithm and transmits the public key to all clients; then, each client encrypts the updated local model parameters w_k^{t+1} using a public key to obtain $\left|w_k^{t+1}\right|$, and uploads $\left|w_k^{t+1}\right|$ to the server; next, the server securely aggregates the encrypted local model parameters from each client, which is computed in Eq. (15); finally, the private key is employed to decrypt $\left|w^{t+1}\right|$ to generate a new round of global model parameters w^{t+1}:

$$\left|w^{t+1}\right| = \sum_{k=1}^{K} weight_k \times \left|w_k^{t+1}\right|. \tag{15}$$

3.5. Detailed modeling steps

To explicitly describe the modeling process of FedCSL-CM, we give the pseudo-code of the proposed method. First, line 1 is the initialization phase of FedCSL-CM, where the server initializes the global model parameters, generates the key pairs for the homomorphic encryption algorithm, and sends the global model parameters and public key to each client. Then, the training process of FedCSL-CM is executed, starting from line 2. Specifically, lines 3–6 execute the parameter update process of local models, where each client computes the probability outcomes of the global model on its local training data and performs the model parameter update ($w_k^{t+1} \leftarrow w_k^t$) by the stochastic gradient descent method. Lines 7–9 contain the learning performance evaluation and parameter encryption process of local models. After the local model parameters are updated, each client uses the performance metrics U_j to comprehensively assess the learning performance of the local model for the current round and encrypts the local model parameters ($\left|w_k^{t+1}\right| \leftarrow w_k^{t+1}$) using the public key. Line 10 uploads $\left|w_k^{t+1}\right|$ and $U_j(w_k^{t+1})$ to the server. Lines 11–13 contain the local model weighting and secure model aggregation processes, where the server utilizes the cloud model and assigns weights to all local models based on their learning performance results. Lines 14–16 implement a secure model aggregation process, where the server computes the encrypted local parameters and corresponding weights, and decrypts the computed results with a private key to generate a new round of the global model. Line 17 returns the latest global model to each client. Finally, the process from lines 2–17 above is repeated until the global model converges or the set number of communications is reached.

3.6. Security analysis

Following the proposed privacy-preserving decentralized training scheme and the associated encryption module, the FedCSL-CM method can fulfill the criteria for data security.

• *Credit data privacy leakage issues*. Compared with collecting data directly from other financial organizations

Algorithm 1: FedCSL-CM

Input: local training data D_k; initial model parameters w^0; number of clients K; number of global rounds T; learning rate η; number of local epochs E; number of evaluation measures m; focusing parameter γ; weighting factor α; relative weight parameter λ; and temperature parameter τ.
Output: global model w.

Server generates key pairs, initializes global parameters w^1, and broadcasts the public key and w^1 to all clients
for global round $t = 1$ **to** T **do**
  for client $k = 1$ **to** K **do** ▷Parallel
    for each local epoch $e = 1$ **to** E **do**
      $w_k^{t+1} \leftarrow w_k^t - \eta \nabla L_k$
    end for
    $U_j(w_k^{t+1}) \leftarrow$ evaluate the learning performance of w_k^{t+1} using the measures U_j ($j = 1, 2, \ldots, m$)
    $\left|w_k^{t+1}\right| \leftarrow$ encrypt w_k^{t+1} using the public key
  end for
  return $\left|w_k^{t+1}\right|$ and $U_j(w_k^{t+1})$ to the server
  for Server **do**
    receive $\left|w_k^{t+1}\right|$ and $U_j(w_k^{t+1})$
    $weight_k \leftarrow$ calculate each local weight using the $U_j(w_k^{t+1})$
    $\left|w^{t+1}\right| \leftarrow \sum_{k=1}^{K} weight_k \times \left|w_k^{t+1}\right|$
    $w^{t+1} \leftarrow$ decrypt $|w^{t+1}|$ using the private key
  end for
  return w^{t+1} to each client
end for
return global model w

to train credit risk prediction models, this paper has better data security in credit risk prediction utilizing multi-party data. The decentralized credit risk prediction method proposed in this paper does not require all financial organizations to transmit their local customer data to a central node, but only intermediate data (i.e., model parameters are transmitted rather than the data itself). In this study, HFL is adopted as a protocol for financial organizations to collaboratively train the shared credit risk prediction model, and each participant achieves secure data sharing among themselves by contributing the parameters of their local model.

• *Model parameter inference attack issues*. To further ensure the privacy information of clients, we incorporate the Paillier homomorphic encryption algorithm in the FedCSL-CM method. The Paillier algorithm satisfies the standard definition of security for encryption schemes: semantic security, that is, ciphertext indistinguishability under chosen-plaintext attack (IND-CPA), which intuitively means that the ciphertext does not reveal any information in the plaintext. Each client uploads encrypted model parameters for secure aggregation by the server. The co-training uses the ciphertext of local model parameters instead of the plaintext, which can effectively prevent malicious attackers from stealing customers' private information by reverse inference of model parameters.

3.7. Complexity and communication overhead analysis

In the FedCSL-CM approach, the computational complexity for K clients performing local model parameter updates is $O(TKEN)$, and the computational complexity for encrypting the updated local model parameters is $O(TKw_k)$. The computational complexity of the server assigning all local models and performing secure aggregation is $O(TKw_k)$. The computational complexity of the server decrypting the global model parameters is $O(Tw_k)$. T is the global number of training rounds, K is the number of clients, E is the local number of training rounds, N is the number of local samples, and w_k is the number of local parameters. Local parameters encryption, secure model aggregation, and global parameters decryption are the operations with the highest computational overhead; the computational overhead of local parameters update and other operations can be approximated to be ignored,

Table 1
Description of the datasets.

Dataset	Samples	Positive samples	Negative samples	Imbalance ratios	Features	Clients
AU	690	307	383	1.2	42	3
LD	1225	323	902	2.8	28	5
HMEQ	5960	1189	4771	4.01	55	10
TW	30,000	6636	23,364	3.52	91	10
LC	42,534	8419	34,115	4.05	34	15
Give	150,000	10,026	139,974	13.96	10	20

so the overall computational complexity can be viewed as $O(Tw_k\ (2K+1))$.

We analyze the communication overhead of the FedCSL-CM method from the encrypted local parameters upload and the decrypted global parameters broadcast. *Encrypted local parameters upload*: The communication overhead for all clients to send the encrypted local parameters to the server is $O(TK)$. *Decrypted global parameters broadcast*: The communication overhead for the server to broadcast the updated global parameters to each client is $O(TK)$. In combination, the overall communication overhead is $O(2TK)$.

4. Experimental design

4.1. Experimental datasets

To evaluate the effectiveness of the proposed method comprehensively on distributed and unbalanced data, six representative and widely recognized credit datasets were selected for empirical analysis, covering a range from small to large sample sizes and from low to high imbalance ratios. The datasets include Australian (AU), Loan Data (LD), HMEQ, Taiwan (TW), Lending Club (LC), and Give. These datasets contain borrowers' financial records, including credit utilization rate, debt-to-income ratio, income level, loan history, and delinquency behavior, among other variables. Detailed descriptions of these variables are available on the GitHub[1] platform. To remove the influence of varying feature scales, we normalized (continuous features) and one-hot coded (categorical features) the above datasets.

A detailed description of these datasets is presented in Table 1. The positive samples are the default samples in the dataset, while the negative samples are the non-default samples. The imbalance ratio refers to the ratio of majority and minority sample sizes. It is obvious that the larger the imbalance ratio, the higher the degree of label skew distribution. The clients refer to the number of horizontal federated learning participants for different datasets (Bi et al., 2023).

4.2. Performance measures

To evaluate the performance of FedCSL-CM comprehensively, four performance measures and two economic measures were employed. Multi-measure evaluation is a commonly adopted approach in credit risk prediction research (Xiao et al., 2024). Performance measures include the AUC-ROC, AUC-PR, KS, and BS^+ (Roy et al., 2019). Economic measures include cost and profit (Verbraken et al., 2014). Table 2 presents the confusion matrix, which is typically employed as the basis for these performance metrics.

Table 2
Confusion matrix.

		Predicted	
		Positive	Negative
Real	Positive	True positives (TP)	False negatives (FN)
	Negative	False positives (FP)	True negatives (TN)

The AUC-ROC is the area under the ROC curve, the x-axis of which represents the false positive rate ($FP/(FP+TN)$), and the y-axis the true positive rate ($TP/(TP+FN)$). The larger the AUC-ROC value, the better the discriminative performance of the model. The AUC-PR is the area under the PR curve, with the x-axis indicating the recall ($TP/(TP+FN)$ and the y-axis the precision ($TP/(TP+FP)$). Similarly, a larger AUC-PR value indicates a better performance of the model. The KS measures the maximum differences in distribution among the cumulative positive and cumulative negative sample rates, and the larger its value, the better the model's ability to distinguish between positive and negative samples. The formula for BS^+ is

$$BS^+ = \frac{1}{D_{min}} \sum_{j=1}^{D_{min}} (y^i - p^i)^2, \tag{16}$$

where y^i is the true class for the positive sample, and p^i is the prediction probability for the positive sample, D_{min} is the number of positive samples. A smaller BS^+ indicates greater classification performance for positive samples,

$$\text{Cost} = FN \cdot \text{ROI} + FP \cdot l, \tag{17}$$

$$\text{Profit} = TP \cdot \text{ROI} + FN \cdot (-\text{ROI}) + FP \cdot (-l) + TN \cdot 0, \tag{18}$$

where ROI is return on investment, l = EAD $\times$ LGD is the economic loss incurred after borrower default, EAD is the outstanding amount at the time of default, and LGD is the proportion of EAD that cannot be recovered.

4.3. Experimental settings

We simulated an HFL scenario with K clients and one server. In our experiments, each dataset was randomly

[1] https://github.com/wuaoala/FedCSL-CM

and evenly partitioned into K subsets, with each subset assigned as the local data of a client. This randomly disrupted data partitioning approach may result in incomplete label distributions on individual clients, particularly for datasets with significant class imbalance. Moreover, we followed a widely used data partitioning practice in credit risk prediction by allocating 80% of each client's data for training and the remaining 20% for testing (Liu et al., 2023a).

We compared the performance with local learning methods, which meant that each client trained local models only on its private dataset (Bercea et al., 2022). We selected the representative benchmark classifiers logistic regression (LR; Berkson, 1944), support vector machine (SVM; Cortes & Vapnik, 1995), extreme gradient boosting (XGB; Chen & Guestrin, 2016), light gradient boosting machine (LightGBM; Ke et al., 2017), and multi-layer perceptron (MLP; Baesens et al., 2003) as the comparison models for local learning methods. The detailed parameter settings for LR, SVM, XGB, and LightGBM are referenced from the work of Gunnarsson et al. (2021); the number of neurons for MLP was determined according to the input–output balance principle (Srivastava et al., 2014), that is, approximately 75% of the input dimensionality. The specific neuron settings for each dataset were 28, 14, 28, 46, 18, and 6, respectively. The detailed parameter settings for each model are provided in Table A.1 in Appendix A.

To evaluate the superior performance of FedCSL-CM, we also compared its performance with benchmark federated learning methods FedAvg (McMahan et al., 2017), FedProx (Li et al., 2020), SCAFFOLD (Karimireddy et al., 2020), FedNova (Wang et al., 2020), and FedKD (Wu et al., 2022a), which are standard baselines in the field of federated learning. MLP was employed as the training model for all federated learning methods, with the same number of neurons as in the local learning methods described above. In all experiments, all clients participated in each round; the number of global communication rounds was set to 50, the number of training passes for each local update was set as $E = 5$, and the cross-entropy was applied as the local loss. For fair comparisons, all federated learning methods used an SGD optimizer with a constant learning rate of 0.01 and a momentum of 0.9. The detailed parameter settings for FedAvg, FedProx, SCAFFOLD, FedNova, and FedKD are referenced in the corresponding literature; see Table A.1 in Appendix A for specific details. Moreover, we constructed the corresponding imbalance learning methods FedAvg-RUS, FedAvg-ROS, and FedAvg by combining FedAvg with the commonly used RUS (Liu & Tsoumakas, 2020), ROS (Moreo et al., 2016), and SMOTE (Chawla et al., 2002) methods, and compared the performance with those methods.

For our proposed FedCSL-CM method, the γ parameter in the cost-sensitive learning function was set to 0, the α parameter was set to 0.8, which means that the local model training focuses more on a smaller number of positive samples, the τ parameter was set to 1, the mean-squared error (MSE) was used as the distillation loss, and the λ parameter was set to 0.1, which implies that the distillation loss constitutes a small proportion to the total loss. In the CM-based adaptive weighting strategy, AUC-ROC, AUC-PR, KS, and BS^+ were used to estimate the training performance of each local model. The economic measures were set as follows: ROI at 10,000 euros, EAD at 100,000 euros, and LGD at 0.45 euros. All reported experimental results represent the average performance across the test sets of all clients, obtained from ten repeated experiments.

5. Experimental results analysis and discussion

5.1. Performance comparison with local learning methods

We compared the performance of our proposed FedCSL-CM with the local learning methods LR, SVM, XGB, LightGBM, and MLP. Local learning methods mean that each client trains local models only on its private dataset, so the results are the average performance of all local models on their test datasets. As can be seen from the results in Table 3, our proposed method performs better than all local learning methods on most of the performance measures for each dataset. This indicates that the proposed method improves the model performance, which is because our proposed method utilizes data from different clients for joint training and expands the scale of the training data compared to local learning methods. In addition, the worst-performing measure of local learning methods in the four datasets is BS^+, which suggests that the unbalanced class distribution severely hinders the training effectiveness of local learning methods (especially for positive samples). Notably, the best-performing measure of our proposed method in the four datasets is also BS^+.

To examine whether there are statistically significant differences between our proposed method and local learning methods, we performed a non-parametric experiment, Friedman , to rank their overall performance, as presented in Table 4. Following Lessmann et al. (2015), we computed the individual and overall rankings of six methods ($K = 6$) across six datasets ($N = 6$). From the results presented in Table 4, all the rankings of FedCSL-CM are higher than those of local learning methods. Moreover, the statistic for each performance measure is higher than the Friedman critical value (2.60), and the p-values are less than the α value (0.05), so the null hypothesis can be rejected. Hence, at the 95% confidence level, we can draw the conclusion that there are statistically significant differences in performance between our proposed method and local learning methods.

Then we employed the Nemenyi test to compare the performance of FedCSL-CM and local learning methods. The test results are presented in Fig. 2, from which we can derive the following conclusions: (1) For AUC-ROC, FedCSL-CM significantly outperforms LR, SVM, LightGBM, and MLP, but the difference from XGB is not significant. XGB significantly outperforms MLP, and there are no significant differences in performance between the remaining model pairs. (2) In terms of AUC-PR, FedCSL-CM significantly outperforms LR, SVM, LightGBM, and MLP,

Table 3
Performance comparison with local learning methods.

Datasets	Measures	Methods					
		LR	SVM	XGB	LightGBM	MLP	FedCSL-CM
AU	AUC-ROC	0.851(±0.010)	0.854(±0.009)	0.850(±0.003)	0.831(±0.008)	0.846(±0.008)	**0.881(±0.007)**
	AUC-PR	0.852(±0.011)	0.854(±0.007)	0.856(±0.014)	0.852(±0.006)	0.827(±0.007)	**0.871(±0.007)**
	KS	0.685(±0.007)	0.713(±0.006)	0.659(±0.009)	0.659(±0.019)	0.676(±0.006)	**0.720(±0.007)**
	BS^+	0.160(±0.020)	0.154(±0.017)	0.171(±0.011)	0.218(±0.019)	0.198(±0.011)	**0.042(±0.028)**
LD	AUC-ROC	0.579(±0.002)	0.584(±0.004)	0.557(±0.004)	0.563(±0.002)	0.572(±0.010)	**0.604(±0.007)**
	AUC-PR	0.328(±0.012)	0.330(±0.015)	0.342(±0.009)	**0.393(±0.008)**	0.305(±0.006)	0.376(±0.011)
	KS	0.287(±0.016)	0.289(±0.012)	0.217(±0.013)	0.283(±0.007)	0.283(±0.009)	**0.334(±0.015)**
	BS^+	0.534(±0.030)	0.528(±0.019)	0.419(±0.026)	0.526(±0.018)	0.581(±0.016)	**0.175(±0.019)**
HMEQ	AUC-ROC	0.737(±0.003)	0.734(±0.006)	0.743(±0.005)	0.739(±0.004)	0.728(±0.005)	**0.764(±0.006)**
	AUC-PR	0.488(±0.006)	0.462(±0.007)	**0.492(±0.011)**	0.468(±0.012)	0.488(±0.005)	0.466(±0.011)
	KS	0.427(±0.013)	0.429(±0.007)	0.432(±0.015)	0.427(±0.012)	0.426(±0.019)	**0.492(±0.017)**
	BS^+	0.491(±0.016)	0.531(±0.015)	0.437(±0.024)	0.602(±0.030)	0.482(±0.014)	**0.242(±0.011)**
TW	AUC-ROC	0.766(±0.004)	0.710(±0.002)	0.759(±0.005)	0.751(±0.008)	0.735(±0.003)	**0.780(±0.003)**
	AUC-PR	0.542(±0.005)	0.512(±0.009)	0.539(±0.009)	0.536(±0.008)	0.490(±0.006)	**0.568(±0.015)**
	KS	0.428(±0.015)	0.415(±0.016)	0.426(±0.010)	0.435(±0.008)	0.378(±0.012)	**0.452(±0.019)**
	BS^+	0.437(±0.017)	0.486(±0.019)	0.471(±0.011)	0.497(±0.026)	0.450(±0.018)	**0.211(±0.020)**
LC	AUC-ROC	0.744(±0.010)	0.773(±0.003)	0.776(±0.007)	0.771(±0.006)	0.759(±0.007)	**0.849(±0.008)**
	AUC-PR	0.477(±0.007)	0.434(±0.013)	0.439(±0.013)	0.425(±0.007)	0.509(±0.008)	**0.665(±0.015)**
	KS	0.461(±0.009)	0.458(±0.011)	0.522(±0.010)	0.513(±0.009)	0.506(±0.020)	**0.559(±0.018)**
	BS^+	0.566(±0.029)	0.540(±0.022)	0.536(±0.015)	0.578(±0.022)	0.602(±0.012)	**0.184(±0.025)**
Give	AUC-ROC	0.646(±0.007)	0.647(±0.002)	**0.655(±0.007)**	**0.655(±0.004)**	0.648(±0.005)	0.640(±0.009)
	AUC-PR	0.111(±0.008)	0.112(±0.006)	0.111(±0.007)	0.113(±0.009)	0.113(±0.011)	**0.117(±0.010)**
	KS	0.221(±0.019)	0.220(±0.012)	0.209(±0.017)	0.209(±0.006)	**0.234(±0.011)**	**0.234(±0.020)**
	BS^+	0.841(±0.018)	0.839(±0.028)	0.820(±0.023)	0.845(±0.020)	0.849(±0.018)	**0.598(±0.012)**

Note: Each optimal performance measure is boldfaced.

Table 4
Friedman test results for local learning methods.

Methods	Measures				
	AUC-ROC	AUC-PR	KS	BS^+	Overall
LR	7.67	7.67	7.00	7.33	7.42
SVM	7.33	9.33	7.67	7.33	7.92
XGB	6.00	5.33	7.33	5.33	6.00
LightGBM	8.00	7.33	9.67	10.33	8.83
MLP	9.33	8.67	8.00	9.67	8.92
FedCSL-CM	**3.67**	**3.67**	**2.33**	**2.00**	**2.92**
Statistics of the Friedman test	28.95	27.71	29.05	24.95	
P-values	2.37×10^{-5}	4.14×10^{-5}	2.27×10^{-5}	1.42×10^{-4}	

Note: Top-ranked values are boldfaced.

but the difference from XGB is not significant. XGB significantly outperforms SVM and MLP. (3) In terms of KS, FedCSL-CM significantly outperforms the other five local learning methods, and there are no significant differences in performance between the remaining model pairs. (4) In terms of BS^+, FedCSL-CM significantly outperforms the other five local learning methods. XGB significantly outperforms MLP and LightGBM.

The above results demonstrate that FedCSL-CM can securely and effectively leverage multi-source unbalanced data to train shareable credit risk prediction models with good discriminative performance, thereby promoting data sharing and decentralized modeling across various financial organizations.

5.2. Performance comparison with benchmark federated learning methods

We compared the performance of FedCSL-CM with the benchmark federated learning methods FedAvg, FedProx, SCAFFOLD, FedNova, and FedKD. All methods were tested using the same parameter settings on four credit datasets, and the results of the performance comparison are shown in Table 5. We find that FedCSL-CM has superior performance to other federated learning methods over all datasets. It is noteworthy that, for credit datasets with high imbalance ratios, the benchmark federated learning methods perform very poorly on the BS^+ measure. This suggests that heavily skewed class distributions can lead to misleading estimates from federated models, especially for positive samples. We can see that our proposed method achieves a significant improvement on this measure, indicating that our proposed method is effective in solving the class imbalance issue.

Further, the Friedman method was employed to test whether there are statistically significant differences between the proposed method and benchmark federated learning methods, as shown in Table 6. In this experiment, we compared the performance rankings of six methods ($K = 6$) across six datasets ($N = 6$). It can be seen that all the rankings of FedCSL-CM are higher than those of benchmark federated learning methods, the statistic of

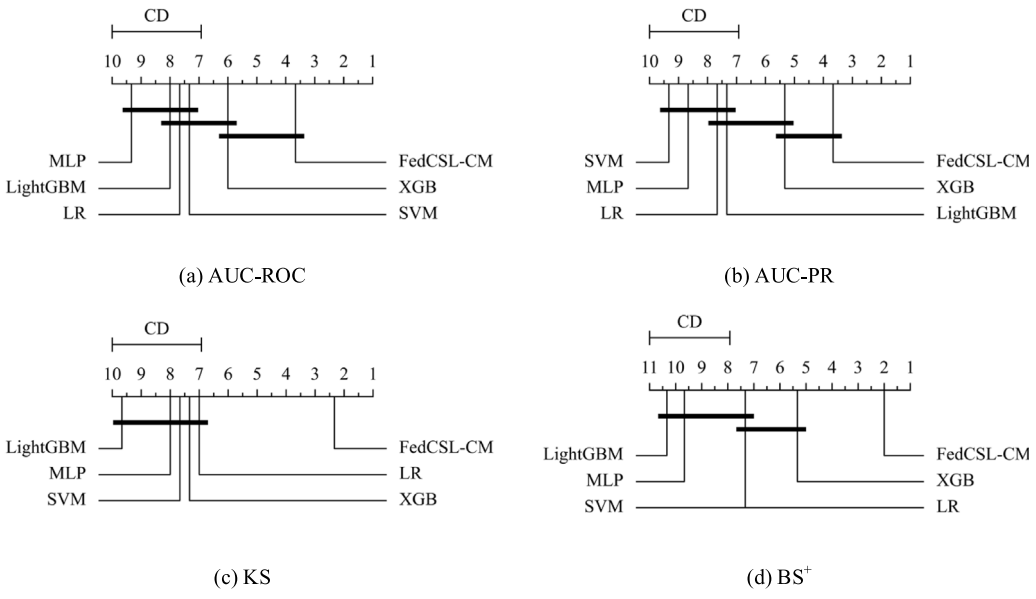


Fig. 2. Nemenyi test results for local learning methods.

Table 5
Performance comparison with benchmark federated learning methods.

Datasets	Measures	Methods					
		FedAvg	FedProx	SCAFFOLD	FedNova	FedKD	FedCSL-CM
AU	AUC-ROC	0.876(±0.009)	0.876(±0.008)	0.879(±0.009)	0.879(±0.009)	0.875(±0.008)	**0.881(±0.007)**
	AUC-PR	0.879(±0.012)	0.879(±0.014)	**0.881(±0.009)**	0.880(±0.006)	0.877(±0.007)	0.871(±0.007)
	KS	0.250(±0.018)	0.709(±0.011)	**0.723(±0.015)**	0.711(±0.013)	0.705(±0.013)	0.720(±0.007)
	BS^+	0.709(±0.013)	0.250(±0.029)	0.251(±0.028)	0.214(±0.010)	0.253(±0.018)	**0.042(±0.028)**
LD	AUC-ROC	0.557(±0.008)	0.559(±0.008)	0.561(±0.003)	0.566(±0.003)	0.535(±0.003)	**0.604(±0.007)**
	AUC-PR	0.339(±0.005)	0.339(±0.006)	0.338(±0.011)	0.358(±0.013)	0.340(±0.007)	**0.376(±0.011)**
	KS	0.267(±0.015)	0.273(±0.019)	0.268(±0.005)	0.292(±0.012)	0.232(±0.011)	**0.334(±0.015)**
	BS^+	0.531(±0.016)	0.531(±0.012)	0.543(±0.013)	0.535(±0.011)	0.522(±0.028)	**0.175(±0.019)**
HMEQ	AUC-ROC	0.739(±0.005)	0.739(±0.008)	0.741(±0.009)	0.752(±0.003)	0.730(±0.007)	**0.764(±0.006)**
	AUC-PR	0.439(±0.013)	0.439(±0.005)	0.441(±0.015)	0.452(±0.012)	0.432(±0.005)	**0.466(±0.011)**
	KS	0.456(±0.018)	0.457(±0.011)	0.456(±0.006)	0.459(±0.017)	0.441(±0.018)	**0.492(±0.017)**
	BS^+	0.587(±0.010)	0.588(±0.021)	0.588(±0.025)	0.518(±0.017)	0.607(±0.011)	**0.242(±0.011)**
TW	AUC-ROC	0.776(±0.006)	0.776(±0.004)	0.774(±0.004)	0.774(±0.010)	0.775(±0.008)	**0.780(±0.003)**
	AUC-PR	0.558(±0.013)	0.558(±0.014)	0.558(±0.010)	0.562(±0.008)	0.555(±0.011)	**0.568(±0.015)**
	KS	0.446(±0.016)	0.446(±0.014)	0.443(±0.005)	0.449(±0.010)	0.445(±0.014)	**0.452(±0.019)**
	BS^+	0.431(±0.027)	0.430(±0.011)	0.438(±0.028)	0.428(±0.012)	0.417(±0.025)	**0.211(±0.020)**
LC	AUC-ROC	0.790(±0.006)	0.789(±0.009)	0.792(±0.004)	0.809(±0.006)	0.778(±0.002)	**0.849(±0.008)**
	AUC-PR	0.529(±0.009)	0.526(±0.013)	0.533(±0.009)	0.590(±0.006)	0.502(±0.005)	**0.665(±0.015)**
	KS	0.479(±0.006)	0.476(±0.005)	0.482(±0.015)	0.496(±0.015)	0.457(±0.006)	**0.559(±0.018)**
	BS^+	0.601(±0.018)	0.602(±0.018)	0.600(±0.026)	0.442(±0.018)	0.607(±0.027)	**0.184(±0.025)**
Give	AUC-ROC	0.634(±0.008)	0.634(±0.009)	0.634(±0.006)	0.622(±0.003)	0.633(±0.005)	**0.640(±0.009)**
	AUC-PR	0.106(±0.013)	0.106(±0.009)	0.106(±0.004)	0.107(±0.013)	0.103(±0.006)	**0.117(±0.010)**
	KS	0.229(±0.010)	0.229(±0.019)	0.229(±0.007)	0.211(±0.018)	0.228(±0.014)	**0.234(±0.020)**
	BS^+	0.864(±0.015)	0.864(±0.029)	0.866(±0.023)	0.832(±0.011)	0.857(±0.029)	**0.598(±0.012)**

Table 6
Friedman test results.

Methods	Measures				
	AUC-ROC	AUC-PR	KS	BS^+	Overall
FedAvg	7.00	7.33	8.33	9.00	7.92
FedProx	8.67	9.00	7.67	7.67	8.25
SCAFFOLD	6.67	7.33	7.00	10.00	7.75
FedNova	6.67	4.00	5.67	5.33	5.42
FedKD	11.00	10.67	11.00	8.00	10.17
FedCSL-CM	**2.00**	**3.67**	**2.33**	**2.00**	**2.50**
Statistics of the Friedman test	30.00	29.24	24.76	21.14	
P-values	1.47×10^{-5}	2.08×10^{-5}	1.55×10^{-4}	7.61×10^{-4}	

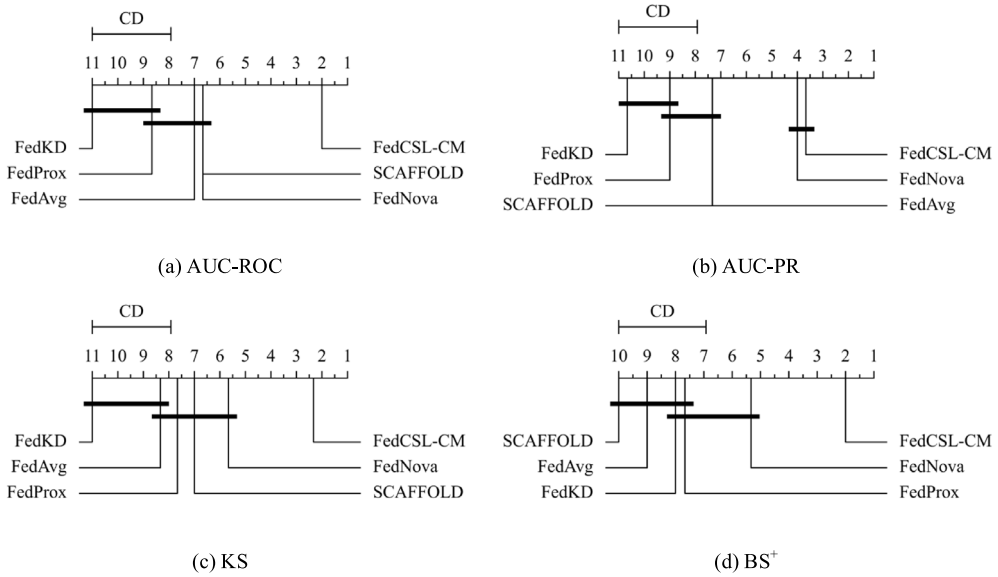


Fig. 3. Nemenyi test results for benchmark federated learning methods.

each performance measure is higher than the Friedman critical value (2.60), and the p-value is less than the α value (0.05), so the null hypothesis is not valid. Hence, at the 95% confidence level, we can draw the conclusion that there are statistically significant differences in performance between our proposed method and the benchmark federated learning method.

Similarly, we conducted the Nemenyi test to further analyze the performance differences between FedCSL-CM and benchmark federated learning methods. Based on the results shown in Fig. 3, we can draw the following conclusions: (1) For AUC-ROC, FedCSL-CM significantly outperforms the other five benchmark federated learning methods. FedKD is significantly inferior to FedAvg, SCAFFOLD, and FedNova, but does not differ significantly from FedProx. There are no significant differences in performance between the remaining model pairs. (2) For AUC-PR, FedCSL-CM significantly performs better than FedAvg, FedProx, SCAFFOLD, and FedKD, but does not differ significantly from FedNova. FedNova significantly outperforms the other four benchmark federated learning methods. (3) In terms of KS, FedCSL-CM significantly outperforms the other five benchmark federated learning methods. FedKD is significantly inferior to FedNova. (4) In terms of BS^{+}, FedCSL-CM significantly outperforms the other five benchmark federated learning methods. FedNova significantly outperforms FedAvg and SCAFFOLD.

In conclusion, the experimental results prove that our proposed method can effectively enhance the performance of federated learning methods and has promising competitiveness for further applications in the context of decentralized credit risk prediction.

5.3. Performance comparison with unbalanced learning methods

To further prove the effectiveness of FedCSL-CM in solving the class imbalance issue, we also compared the performance of FedCSL-CM with those of FedAvg-RUS, FedAvg-ROS, and FedAvg-SMOTE. The performance comparison results are given in Table 7, from which it can be seen that our proposed method obtains the optimal performance most times on all datasets. Among them, RUS, a method that balances the original dataset by deleting the majority of class samples, performs the worst on most datasets, probably because it causes more serious information loss by deleting the training samples in decentralized scenarios that require shared data, which affects the training effect of local models. The two methods of generating minority class samples, ROS and SMOTE, also fail to achieve the optimal state, probably because the samples they generate do not accurately reflect the real default situation, which affects the reality of the experiment.

The above experimental results show that resampling local training samples without allowing global analysis may cause information loss or generate additional noisy information, thus impairing the training effect of decentralized modeling. In contrast, FedCSL-CM utilizes the existing private credit data of each client to obtain superior performance without removing local majority class samples or generating more minority class samples. This suggests that FedCSL-CM is better suitable to handling unbalanced data distributions in decentralized credit risk prediction scenarios.

Table 7
Performance comparison with unbalanced learning methods.

Datasets	Measures	Methods			
		FedAvg-RUS	FedAvg-ROS	FedAvg-SMOTE	FedCSL-CM
AU	AUC-ROC	0.877(±0.006)	0.878(±0.005)	0.875(±0.007)	**0.881(±0.007)**
	AUC-PR	0.875(±0.009)	**0.887(±0.006)**	0.875(±0.008)	0.871(±0.007)
	KS	0.708(±0.011)	**0.737(±0.008)**	0.721(±0.010)	0.720(±0.007)
	BS^+	0.193(±0.015)	0.175(±0.012)	0.175(±0.014)	**0.042(±0.028)**
LD	AUC-ROC	0.564(±0.012)	0.592(±0.010)	0.588(±0.011)	**0.604(±0.007)**
	AUC-PR	0.327(±0.015)	0.367(±0.014)	0.353(±0.013)	**0.376(±0.011)**
	KS	0.262(±0.018)	0.264(±0.016)	0.257(±0.019)	**0.334(±0.015)**
	BS^+	0.267(±0.022)	0.256(±0.020)	0.261(±0.021)	**0.175(±0.019)**
HMEQ	AUC-ROC	0.709(±0.008)	0.761(±0.007)	0.743(±0.008)	**0.764(±0.006)**
	AUC-PR	0.395(±0.012)	0.458(±0.011)	0.436(±0.013)	**0.466(±0.011)**
	KS	0.418(±0.014)	0.480(±0.012)	0.475(±0.014)	**0.492(±0.017)**
	BS^+	0.241(±0.018)	0.225(±0.016)	**0.223(±0.014)**	0.242(±0.011)
TW	AUC-ROC	0.772(±0.005)	0.780(±0.004)	0.776(±0.005)	**0.780(±0.003)**
	AUC-PR	0.538(±0.009)	0.559(±0.008)	0.555(±0.009)	**0.568(±0.015)**
	KS	0.445(±0.011)	0.451(±0.010)	0.448(±0.012)	**0.452(±0.019)**
	BS^+	**0.197(±0.012)**	0.199(±0.013)	0.201(±0.014)	**0.211(±0.020)**
LC	AUC-ROC	0.766(±0.007)	0.840(±0.006)	0.840(±0.005)	**0.849(±0.008)**
	AUC-PR	0.466(±0.014)	0.645(±0.010)	0.647(±0.011)	**0.665(±0.015)**
	KS	0.441(±0.016)	0.551(±0.012)	0.550(±0.013)	**0.559(±0.018)**
	BS^+	0.245(±0.024)	0.195(±0.020)	0.195(±0.021)	**0.184(±0.025)**
Give	AUC-ROC	0.624(±0.010)	0.633(±0.009)	0.633(±0.008)	**0.640(±0.009)**
	AUC-PR	0.108(±0.012)	**0.119(±0.010)**	0.119(±0.011)	0.117(±0.010)
	KS	0.212(±0.018)	0.234(±0.015)	**0.235(±0.014)**	0.234(±0.020)
	BS^+	**0.247(±0.021)**	0.249(±0.023)	0.249(±0.022)	0.598(±0.012)

Table 8
Results of ablation experiments.

Datasets	Measures	Methods			
		Baseline	FedCSL	FedCM	FedCSL-CM
AU	AUC-ROC	0.792(±0.008)	0.807(±0.006)	0.805(±0.007)	**0.815(±0.005)**
	AUC-PR	0.801(±0.009)	0.810(±0.008)	0.810(±0.008)	**0.819(±0.006)**
	KS	0.590(±0.012)	**0.607(±0.010)**	0.597(±0.011)	0.603(±0.009)
	BS^+	0.251(±0.018)	0.106(±0.012)	0.202(±0.015)	**0.091(±0.008)**
LD	AUC-ROC	0.519(±0.014)	0.575(±0.011)	0.570(±0.012)	**0.577(±0.009)**
	AUC-PR	0.324(±0.015)	0.345(±0.013)	0.339(±0.014)	**0.350(±0.011)**
	KS	0.238(±0.018)	0.280(±0.016)	0.291(±0.016)	**0.299(±0.014)**
	BS^+	0.486(±0.025)	0.209(±0.019)	0.491(±0.026)	**0.193(±0.015)**
HMEQ	AUC-ROC	0.700(±0.009)	0.711(±0.007)	0.708(±0.008)	**0.720(±0.006)**
	AUC-PR	0.397(±0.014)	0.414(±0.012)	0.411(±0.013)	**0.414(±0.010)**
	KS	0.392(±0.015)	0.430(±0.013)	0.424(±0.014)	**0.427(±0.011)**
	BS^+	0.563(±0.028)	0.242(±0.018)	0.509(±0.022)	**0.240(±0.016)**
TW	AUC-ROC	0.764(±0.006)	0.766(±0.005)	0.769(±0.005)	**0.775(±0.004)**
	AUC-PR	0.526(±0.011)	0.545(±0.009)	0.549(±0.010)	**0.554(±0.008)**
	KS	0.429(±0.014)	0.444(±0.012)	0.442(±0.013)	**0.445(±0.010)**
	BS^+	0.463(±0.024)	**0.195(±0.015)**	0.421(±0.022)	0.197(±0.014)
LC	AUC-ROC	0.691(±0.010)	0.771(±0.007)	0.741(±0.008)	**0.782(±0.005)**
	AUC-PR	0.383(±0.015)	0.516(±0.011)	0.460(±0.013)	**0.528(±0.009)**
	KS	0.322(±0.017)	0.449(±0.014)	0.437(±0.015)	**0.461(±0.012)**
	BS^+	0.597(±0.031)	0.227(±0.018)	0.458(±0.025)	**0.226(±0.016)**
Give	AUC-ROC	0.633(±0.011)	0.636(±0.009)	0.637(±0.009)	**0.637(±0.008)**
	AUC-PR	0.104(±0.013)	0.113(±0.011)	0.111(±0.012)	**0.113(±0.010)**
	KS	0.227(±0.016)	0.231(±0.014)	**0.233(±0.014)**	0.231(±0.013)
	BS^+	0.778(±0.035)	0.573(±0.025)	0.859(±0.040)	**0.572(±0.022)**

5.4. Ablation analysis

To illustrate the effectiveness of our proposed method, we evaluated the importance of the proposed modules for the performance improvement of FedCSL-CM. We employed the FedAvg method as a baseline for this ablation experiment. Table 8 presents the average results of all methods across all communication rounds. As can be seen from Table 8, the introduction of the CSL-based local model parameter update method and the CM-based adaptive local model weighting strategy separately during the training process consistently improved the model performance on most measures.

Furthermore, we find that our proposed final FedCSL-CM has the best performance compared with applying only one module. Overall, these results indicate that in a decentralized credit risk prediction scenario, our proposed CSL-based local model parameter update method demonstrates superior adaptability to unbalanced class distributions, while the CM-based adaptive local model

Table 9
Performance comparison of different test sample ratios.

Datasets	Methods	Test sample ratios				
		50%	40%	30%	20%	10%
AU	FedAvg	0.787(±0.008)	0.788(±0.007)	0.799(±0.006)	0.792(±0.008)	0.793(±0.009)
	FedProx	0.786(±0.007)	0.788(±0.007)	0.799(±0.006)	0.792(±0.008)	0.793(±0.009)
	SCAFFOLD	0.787(±0.008)	0.789(±0.007)	0.800(±0.007)	0.794(±0.008)	0.794(±0.010)
	FedNova	0.789(±0.007)	0.790(±0.007)	0.801(±0.006)	0.797(±0.007)	0.801(±0.008)
	FedKD	0.776(±0.010)	0.777(±0.010)	0.790(±0.009)	0.784(±0.011)	0.781(±0.012)
	FedCSL-CM	**0.809(±0.005)**	**0.807(±0.006)**	**0.814(±0.004)**	**0.815(±0.005)**	**0.822(±0.006)**
LD	FedAvg	0.543(±0.012)	0.560(±0.011)	0.545(±0.013)	0.519(±0.014)	0.525(±0.015)
	FedProx	0.543(±0.012)	0.558(±0.011)	0.546(±0.012)	0.518(±0.014)	0.524(±0.016)
	SCAFFOLD	0.544(±0.013)	0.562(±0.012)	0.546(±0.013)	0.521(±0.014)	0.520(±0.017)
	FedNova	0.542(±0.012)	0.558(±0.011)	0.549(±0.011)	0.525(±0.013)	0.525(±0.015)
	FedKD	0.543(±0.014)	0.553(±0.013)	0.540(±0.014)	0.514(±0.015)	0.514(±0.018)
	FedCSL-CM	**0.566(±0.008)**	**0.603(±0.007)**	**0.601(±0.007)**	**0.577(±0.009)**	**0.638(±0.010)**
HMEQ	FedAvg	0.702(±0.009)	0.704(±0.008)	0.719(±0.007)	0.700(±0.009)	0.718(±0.011)
	FedProx	0.702(±0.009)	0.704(±0.008)	0.719(±0.007)	0.699(±0.010)	0.717(±0.011)
	SCAFFOLD	0.703(±0.010)	0.705(±0.009)	0.720(±0.008)	0.701(±0.010)	0.719(±0.012)
	FedNova	0.694(±0.011)	0.699(±0.010)	0.717(±0.009)	0.700(±0.011)	0.715(±0.012)
	FedKD	0.697(±0.012)	0.699(±0.011)	0.715(±0.010)	0.696(±0.012)	0.713(±0.014)
	FedCSL-CM	**0.712(±0.006)**	**0.719(±0.006)**	**0.740(±0.005)**	**0.720(±0.006)**	**0.756(±0.008)**
TW	FedAvg	0.748(±0.007)	0.752(±0.006)	0.757(±0.006)	0.764(±0.006)	0.778(±0.007)
	FedProx	0.748(±0.007)	0.752(±0.006)	0.757(±0.006)	0.764(±0.005)	0.778(±0.007)
	SCAFFOLD	0.749(±0.008)	0.753(±0.007)	0.758(±0.007)	0.764(±0.006)	0.777(±0.008)
	FedNova	0.743(±0.009)	0.747(±0.008)	0.753(±0.008)	0.760(±0.007)	0.775(±0.009)
	FedKD	0.747(±0.010)	0.750(±0.009)	0.755(±0.009)	0.763(±0.008)	0.777(±0.011)
	FedCSL-CM	**0.762(±0.004)**	**0.764(±0.004)**	**0.768(±0.004)**	**0.775(±0.004)**	**0.788(±0.005)**
LC	FedAvg	0.666(±0.011)	0.666(±0.011)	0.692(±0.010)	0.691(±0.010)	0.686(±0.013)
	FedProx	0.665(±0.011)	0.665(±0.011)	0.691(±0.010)	0.690(±0.010)	0.686(±0.013)
	SCAFFOLD	0.667(±0.012)	0.667(±0.012)	0.694(±0.011)	0.692(±0.011)	0.688(±0.014)
	FedNova	0.699(±0.009)	0.701(±0.008)	0.725(±0.007)	0.721(±0.008)	0.711(±0.010)
	FedKD	0.656(±0.014)	0.656(±0.014)	0.682(±0.013)	0.680(±0.013)	0.676(±0.016)
	FedCSL-CM	**0.751(±0.007)**	**0.754(±0.006)**	**0.784(±0.005)**	**0.782(±0.005)**	**0.776(±0.008)**
Give	FedAvg	0.635(±0.010)	0.638(±0.009)	0.633(±0.010)	0.633(±0.011)	0.630(±0.012)
	FedProx	0.635(±0.010)	0.638(±0.009)	0.633(±0.010)	0.633(±0.010)	0.630(±0.012)
	SCAFFOLD	0.635(±0.011)	0.638(±0.010)	0.633(±0.011)	0.633(±0.011)	0.630(±0.013)
	FedNova	0.628(±0.012)	0.629(±0.011)	0.623(±0.012)	0.622(±0.012)	0.623(±0.014)
	FedKD	0.635(±0.013)	0.638(±0.012)	0.633(±0.013)	0.633(±0.013)	0.629(±0.015)
	FedCSL-CM	**0.638(±0.007)**	**0.642(±0.006)**	**0.638(±0.007)**	**0.637(±0.008)**	**0.636(±0.009)**

weighting strategy can comprehensively evaluate the learning performance of each local model and assign weights reasonably.

5.5. Robustness analysis

In this subsection, we adjust the ratio between local training samples and test samples to examine the robustness of the proposed method. We selected the AUC-ROC as a representative performance measure, and the results are shown in Table 9, where {50%, 40%, 30%, 20%, 10%} denotes the proportion of test samples in the local dataset, which represents different local training sample scales. From the results given in Table 9, we obtain the following findings: (1) Our proposed FedCSL-CM outperforms benchmark federated learning methods on all datasets. (2) The performance of FedCSL-CM is robust on different local training sample scales. (3) Among benchmark federated learning methods, SCAFFOLD performs best and outperforms other benchmark methods, while FedKD performs worst. (4) Increasing the training samples can significantly enhance the predictive performance of the model.

Therefore, the above findings show that our proposed FedCSL-CM can effectively improve credit risk prediction with good robustness.

5.6. Training time analysis

In this subsection, we compare the computational time of FedCSL-CM with that of the benchmark federated learning methods. Fig. 4 illustrates the training time of the compared methods on each dataset. It can be observed that FedCSL-CM requires more training time than the other methods.

This is mainly due to the following reasons: (1) *CSL-based local model parameter updates*: The loss function of FedCSL-CM consists of a local focal loss term and a global distillation loss term. During local model updates, each client must not only compute the cost information of local unbalanced training samples but also calculate the probabilistic outputs from the global model. This additional computation increases the complexity of local model updates, thereby leading to higher training time overhead. (2) CM-based adaptive local model weighting: After each round of local model updates, FedCSL-CM evaluates the overall learning performance of each local model in the current round and assigns corresponding weights based on this performance. Compared with direct averaging used in other benchmark federated learning methods, the CM-based adaptive local model weighting strategy introduces additional computational complexity. (3) HE-based secure model aggregation: The training process of

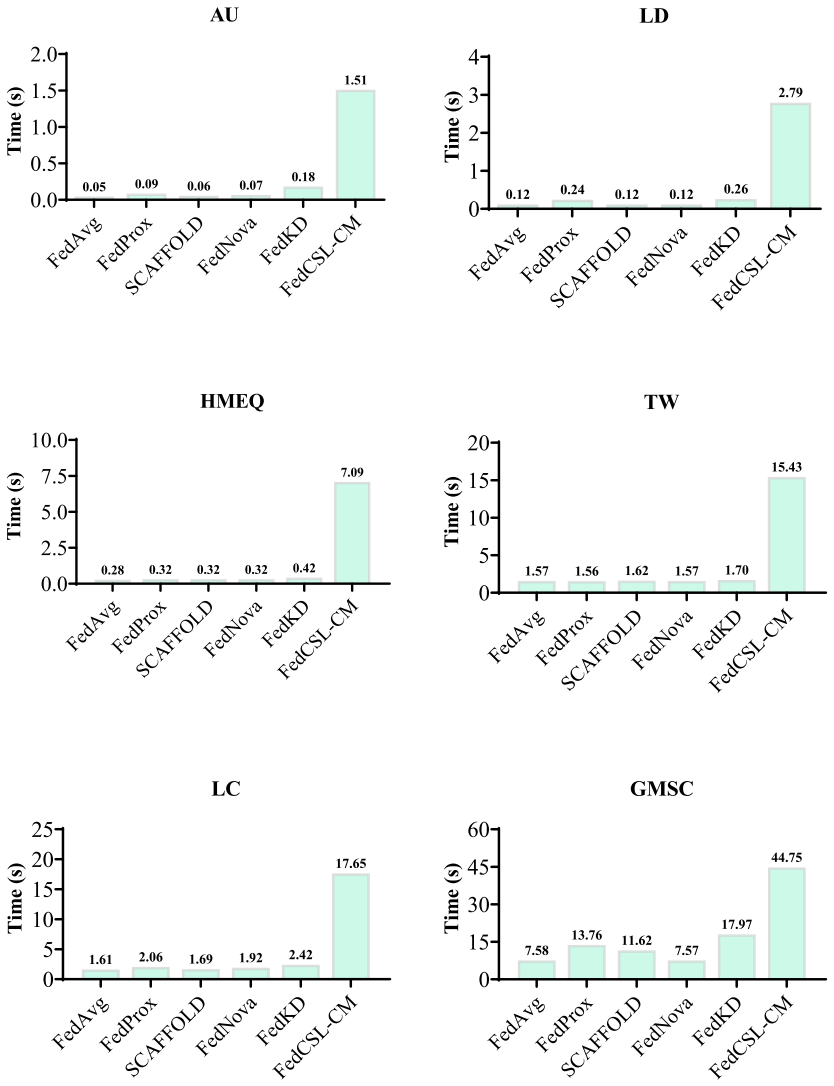


Fig. 4. Time consumption of different federated learning methods.

FedCSL-CM integrates a homomorphic encryption algorithm to ensure secure model aggregation, which involves both local parameter encryption and global parameter decryption. Specifically, during local updates, clients first encrypt their locally updated model parameters before uploading them to the server. During global aggregation, the server computes the encrypted local parameters and their corresponding weights, then decrypts the aggregated results to obtain the global model parameters. Consequently, additional computational time is required to perform these encryption and decryption operations throughout the training process.

5.7. Key parameter sensitivity analysis

In this subsection, we conduct sensitivity tests on FedCSL-CM for key parameters. The key parameters of our proposed FedCSL-CM are the focusing parameter γ, the weighting parameter α, and the relative weight parameter λ of the cost-sensitive learning function. In principle, the higher the γ value, the more the model training focuses on hard examples; the higher the α value, the more the model training focuses on minority class samples; and the higher the λ value, the more the model training focuses on the distillation loss. We first tested the performance of FedCSL-CM for different values of the focusing parameter γ selected from {0, 1, 2, 5} when λ is set to 0.1 and α is set to 0.8. From the average performance results reported in Table 10, it can be observed that FedCSL-CM performs best on most of the measures when γ is set to 0. Furthermore, it is worth noting that the performance achieved with γ = 5 is the worst, which could potentially be because FedCSL-CM overly reduces the loss contribution of easy examples, thus compromising the overall performance of the model.

We then compared the effect of different values of the weighting parameter α selected from {0.6, 0.7, 0.8, 0.9} on FedCSL-CM when γ was set to 0 and λ was set to 0.1, as reported in Table 11. We find that FedCSL-CM performs

Table 10
Performance results of FedCSL-CM with different values of γ.

Datasets	Measures	Parameters			
		$\gamma = 0$	$\gamma = 1$	$\gamma = 2$	$\gamma = 5$
AU	AUC-ROC	**0.815(±0.005)**	0.795(±0.007)	0.758(±0.008)	0.553(±0.015)
	AUC-PR	**0.819(±0.006)**	0.805(±0.008)	0.772(±0.010)	0.596(±0.014)
	KS	0.603(±0.009)	0.578(±0.013)	0.532(±0.015)	0.270(±0.022)
	BS^+	**0.091(±0.008)**	0.132(±0.012)	0.164(±0.015)	0.209(±0.018)
LD	AUC-ROC	**0.577(±0.009)**	0.566(±0.010)	0.551(±0.012)	0.506(±0.016)
	AUC-PR	**0.350(±0.011)**	0.344(±0.012)	0.330(±0.014)	0.280(±0.018)
	KS	**0.299(±0.014)**	0.276(±0.016)	0.253(±0.017)	0.240(±0.020)
	BS^+	**0.193(±0.015)**	0.219(±0.018)	0.236(±0.021)	0.267(±0.025)
HMEQ	AUC-ROC	**0.720(±0.006)**	0.717(±0.007)	0.702(±0.009)	0.629(±0.012)
	AUC-PR	**0.414(±0.010)**	0.411(±0.011)	0.396(±0.013)	0.322(±0.016)
	KS	**0.427(±0.011)**	0.421(±0.012)	0.399(±0.014)	0.284(±0.019)
	BS^+	0.240(±0.016)	**0.237(±0.014)**	0.239(±0.017)	0.244(±0.021)
TW	AUC-ROC	**0.775(±0.004)**	0.774(±0.005)	0.772(±0.005)	0.746(±0.008)
	AUC-PR	**0.554(±0.008)**	0.551(±0.009)	0.546(±0.010)	0.497(±0.013)
	KS	**0.445(±0.010)**	0.444(±0.011)	0.437(±0.012)	0.404(±0.015)
	BS^+	0.197(±0.014)	**0.194(±0.011)**	0.203(±0.013)	0.226(±0.016)
LC	AUC-ROC	**0.782(±0.005)**	0.773(±0.006)	0.748(±0.008)	0.624(±0.015)
	AUC-PR	**0.528(±0.009)**	0.510(±0.011)	0.461(±0.013)	0.283(±0.018)
	KS	**0.461(±0.012)**	0.448(±0.014)	0.412(±0.016)	0.232(±0.022)
	BS^+	**0.226(±0.016)**	0.233(±0.016)	0.241(±0.017)	0.248(±0.019)
Give	AUC-ROC	**0.637(±0.008)**	0.637(±0.008)	0.634(±0.009)	0.627(±0.011)
	AUC-PR	0.113(±0.010)	**0.114(±0.010)**	0.112(±0.011)	0.108(±0.012)
	KS	**0.231(±0.013)**	0.228(±0.015)	0.225(±0.015)	0.216(±0.017)
	BS^+	0.572(±0.022)	0.434(±0.020)	0.369(±0.018)	**0.299(±0.015)**

best on most measures when α is set to 0.8. When $\alpha = 0.6$, it performs the least well overall, which suggests that FedCSL-CM should focus more on the learning of minority class samples. In addition, with larger and larger values of α, FedCSL-CM performs better and better on the BS^+ measure but is not optimal on other measures, which suggests that the proposed method should not be overly biased towards minority class samples at the expense of the overall discriminative performance.

Finally, we compared the effect of different values of the relative weight parameter λ selected from {0.1, 0.3, 0.5, 0.7, 0.9} on FedCSL-CM when γ is set to 0 and α is set to 0.8, as reported in Table 12. We find that FedCSL-CM provides the optimal performance on most measures when λ is set to 0.1. This is consistent with the intended setup of previous experiments. In addition, the performance of FedCSL-CM becomes increasingly worse (except for the BS^+ measure) when the λ value increases, suggesting that the distillation term should not affect the local model update unduly.

Taken together, the above results indicate that the parameter configuration used in this study achieves statistically optimal classification performance. Moreover, the experimental findings reflect the practical business logic of financial institutions, which prioritize identifying potential default customers and mitigating bad debt risks in real lending decisions. In other words, under unbalanced credit data distributions, the credit risk prediction model imposes stronger penalties on the minority class (default samples), aligning more closely with the risk preferences of financial institutions.

5.8. Economic analysis

In this subsection, we conduct a comparative analysis of the proposed method and all baseline approaches in terms of economic benefits. We compare these methods using cost and profit measures on the real-world dataset LC, taking the experimental results of FedAvg as the baseline. As shown in Table 13, we can observe that the local learning methods (LR, SVM, XGBoost, LightGBM, and MLP) generally incur higher operational costs, with profits remaining negative. This indicates that independently trained predictive models based on limited data from a single institution struggled to reduce operational costs effectively or achieve meaningful economic gains in practical business scenarios. The benchmark federated learning methods (FedAvg, FedProx, SCAFFOLD, FedNova, and FedKD) are able to integrate cross-institutional data and mitigate data silo issues. However, since they fail to address the class imbalance inherent in decentralized data, their predictive outcomes do not demonstrate a clear economic advantage—profits remain negative or near zero. This suggests that traditional parameter aggregation strategies alone are insufficient to translate improvements in predictive performance into tangible business value.

The federated methods incorporating unbalanced learning strategies (FedAvg-RUS, FedAvg-ROS, and FedAvg-SMOTE) achieved positive profits, demonstrating an improvement in economic performance. This indicates that employing RUS, ROS, or SMOTE enhanced the model's ability to identify high-risk minority samples,

Table 11
Performance results of FedCSL-CM with different values of α.

Datasets	Measures	Parameters			
		$\alpha = 0.6$	$\alpha = 0.7$	$\alpha = 0.8$	$\alpha = 0.9$
AU	AUC-ROC	0.802(±0.007)	0.805(±0.006)	**0.815(±0.005)**	0.808(±0.007)
	AUC-PR	0.811(±0.009)	0.811(±0.008)	**0.819(±0.006)**	0.808(±0.008)
	KS	0.614(±0.013)	**0.615(±0.011)**	0.603(±0.009)	0.586(±0.015)
	BS^+	0.177(±0.015)	0.130(±0.012)	0.091(±0.008)	**0.061(±0.005)**
LD	AUC-ROC	0.551(±0.012)	0.560(±0.011)	**0.577(±0.009)**	0.574(±0.009)
	AUC-PR	0.341(±0.014)	0.346(±0.013)	**0.350(±0.011)**	0.338(±0.013)
	KS	0.264(±0.017)	0.268(±0.016)	**0.299(±0.014)**	0.296(±0.015)
	BS^+	0.393(±0.025)	0.292(±0.021)	0.193(±0.015)	**0.102(±0.009)**
HMEQ	AUC-ROC	**0.734(±0.007)**	0.730(±0.008)	0.720(±0.006)	0.693(±0.012)
	AUC-PR	0.404(±0.012)	0.406(±0.011)	**0.414(±0.010)**	0.386(±0.014)
	KS	0.427(±0.014)	0.422(±0.015)	**0.427(±0.011)**	0.381(±0.017)
	BS^+	0.456(±0.028)	0.354(±0.024)	0.240(±0.016)	**0.108(±0.011)**
TW	AUC-ROC	0.766(±0.006)	0.771(±0.005)	**0.775(±0.004)**	0.771(±0.005)
	AUC-PR	0.548(±0.010)	0.547(±0.009)	**0.554(±0.008)**	0.544(±0.009)
	KS	0.446(±0.012)	**0.447(±0.011)**	0.445(±0.010)	0.439(±0.013)
	BS^+	0.360(±0.022)	0.285(±0.019)	0.197(±0.014)	**0.094(±0.008)**
LC	AUC-ROC	0.779(±0.006)	0.779(±0.005)	**0.782(±0.005)**	0.744(±0.008)
	AUC-PR	0.511(±0.011)	0.517(±0.010)	**0.528(±0.009)**	0.472(±0.014)
	KS	0.451(±0.012)	0.451(±0.011)	**0.461(±0.012)**	0.405(±0.015)
	BS^+	0.454(±0.025)	0.344(±0.018)	0.226(±0.016)	**0.097(±0.009)**
Give	AUC-ROC	0.636(±0.011)	0.637(±0.008)	**0.637(±0.008)**	0.635(±0.011)
	AUC-PR	0.109(±0.013)	0.111(±0.011)	**0.113(±0.010)**	0.113(±0.012)
	KS	0.230(±0.016)	0.230(±0.015)	**0.231(±0.013)**	0.226(±0.017)
	BS^+	0.785(±0.035)	0.701(±0.030)	0.572(±0.022)	**0.351(±0.018)**

Table 12
Performance results of FedCSL-CM with different values of λ.

Datasets	Measures	Parameters				
		$\lambda = 0.1$	$\lambda = 0.3$	$\lambda = 0.5$	$\lambda = 0.7$	$\lambda = 0.9$
AU	AUC-ROC	**0.815(±0.005)**	0.798(±0.007)	0.769(±0.008)	0.717(±0.010)	0.586(±0.015)
	AUC-PR	**0.819(±0.006)**	0.805(±0.008)	0.781(±0.009)	0.731(±0.011)	0.619(±0.014)
	KS	**0.603(±0.009)**	0.573(±0.013)	0.527(±0.014)	0.451(±0.016)	0.278(±0.021)
	BS^+	**0.091(±0.008)**	0.104(±0.009)	0.122(±0.011)	0.148(±0.013)	0.185(±0.015)
LD	AUC-ROC	**0.577(±0.009)**	0.569(±0.010)	0.558(±0.011)	0.537(±0.012)	0.499(±0.014)
	AUC-PR	**0.350(±0.011)**	0.342(±0.012)	0.328(±0.013)	0.299(±0.015)	0.274(±0.017)
	KS	**0.299(±0.014)**	0.290(±0.015)	0.271(±0.016)	0.247(±0.018)	0.236(±0.019)
	BS^+	**0.193(±0.015)**	0.199(±0.016)	0.212(±0.018)	0.237(±0.020)	0.286(±0.023)
HMEQ	AUC-ROC	0.720(±0.006)	0.709(±0.008)	0.693(±0.010)	0.672(±0.011)	0.631(±0.013)
	AUC-PR	**0.414(±0.010)**	0.401(±0.012)	0.385(±0.014)	0.361(±0.015)	0.321(±0.018)
	KS	**0.427(±0.011)**	0.410(±0.013)	0.385(±0.015)	0.354(±0.017)	0.291(±0.020)
	BS^+	0.240(±0.016)	0.239(±0.015)	0.238(±0.015)	0.237(±0.014)	**0.234(±0.012)**
TW	AUC-ROC	**0.775(±0.004)**	0.774(±0.005)	0.772(±0.005)	0.768(±0.006)	0.758(±0.008)
	AUC-PR	**0.554(±0.008)**	0.550(±0.009)	0.543(±0.010)	0.531(±0.011)	0.503(±0.013)
	KS	**0.445(±0.010)**	0.443(±0.011)	0.440(±0.011)	0.435(±0.012)	0.417(±0.015)
	BS^+	0.197(±0.014)	0.194(±0.011)	0.191(±0.011)	0.185(±0.010)	**0.180(±0.009)**
LC	AUC-ROC	**0.782(±0.005)**	0.766(±0.007)	0.742(±0.008)	0.700(±0.011)	0.618(±0.016)
	AUC-PR	**0.528(±0.009)**	0.494(±0.011)	0.448(±0.013)	0.378(±0.016)	0.278(±0.020)
	KS	**0.461(±0.012)**	0.436(±0.014)	0.400(±0.015)	0.338(±0.018)	0.215(±0.024)
	BS^+	**0.226(±0.016)**	0.235(±0.016)	0.242(±0.017)	0.248(±0.019)	0.251(±0.021)
Give	AUC-ROC	**0.637(±0.008)**	0.637(±0.008)	0.636(±0.009)	0.635(±0.010)	0.633(±0.011)
	AUC-PR	**0.113(±0.010)**	0.112(±0.011)	0.111(±0.012)	0.110(±0.012)	0.109(±0.013)
	KS	**0.231(±0.013)**	0.230(±0.015)	0.229(±0.015)	0.228(±0.016)	0.224(±0.018)
	BS^+	0.572(±0.022)	0.564(±0.024)	0.549(±0.022)	0.517(±0.020)	**0.422(±0.017)**

thereby improving business returns to some extent. However, their operational costs remained relatively high, which limits the overall profitability. In contrast, the proposed FedCSL-CM model achieves the best performance in both cost control and profit enhancement. This result demonstrates that FedCSL-CM not only improves predictive accuracy but also translates its performance advantages into practical economic benefits. In particular, within decentralized credit risk prediction tasks, the proposed method simultaneously optimizes predictive accuracy and class imbalance handling, significantly improves the identification of high-risk clients, thereby

Table 13
Economic comparison results.

Methods	Models	Cost (euros)	Profit (euros)
Local learning methods	LR	10832.59(±124.5)	−1158.59(±45.2)
	SVM	10808.37(±98.2)	−1125.55(±38.7)
	XGB	10607.92(±156.4)	−1224.66(±52.1)
	LightGBM	8920.70(±85.6)	−903.08(±31.4)
	MLP	10878.85(±112.0)	−505.06(±25.8)
Benchmark federated learning methods	FedAvg	8920.70(±95.4)	−903.08(±34.2)
	FedProx	8920.70(±92.1)	−903.08(±33.5)
	SCAFFOLD	8920.70(±88.7)	−903.08(±30.9)
	FedNova	5359.03(±62.5)	1064.70(±22.4)
	FedKD	8920.70(±105.3)	−903.08(±36.8)
Unbalanced learning methods	FedAvg-RUS	5193.83(±58.2)	1198.23(±19.6)
	FedAvg-ROS	4648.31(±45.9)	1586.63(±15.4)
	FedAvg-SMOTE	4560.93(±42.7)	1688.69(±14.2)
Our proposed method	FedCSL-CM	**4034.50(±32.4)**	**2281.20(±12.8)**

reducing potential default costs, and enhances overall profitability.

From a business perspective, this finding provides important insights: In multi-institutional financial collaborations, the proposed FedCSL-CM balances privacy protection, predictive performance, and actual profitability, demonstrating strong potential for practical deployment.

5.9. Discussion

Four main aspects of the core contributions of FedCSL-CM can be summarized. First, it introduces an HFL framework into decentralized credit risk prediction, enabling the secure utilization of multi-party data samples. Second, it proposes a CSL-based local model parameter updating method, allowing the proposed framework to better adapt to unbalanced data distributions. Third, it develops a CM-based adaptive local model weighting strategy to mitigate the problem of local model parameter drift caused by data imbalance. Fourth, it integrates homomorphic encryption into the HFL framework, further enhancing the security of the proposed method.

Moreover, the experimental results show that the proposed method handles unbalanced and decentralized credit data effectively, achieves strong discriminative performance, and demonstrates superior cost-saving capabilities while improving overall profitability.

Finally, this study offers several managerial implications that can help financial organizations control credit risk more effectively: (1) Although the financial industry generates a large amount of data that can be applied to construct credit risk prediction models, the use of such data is limited by various privacy regulations, thus creating data silos. Our proposed method enables decentralized credit risk prediction leveraging dispersed data from multiple sources without compromising user privacy. This implies that our method has potential value for further application in facilitating large-scale collaboration among financial organizations. (2) For federated models that transmit local parameters directly to a central server, the malicious client and server are still able to obtain other clients' private data information through reverse inference engineering. This suggests that traditional federated frameworks may not be completely secure for data privacy. Financial organizations need to further consider model security when implementing decentralized modeling. (3) Imbalanced credit data pose a significant challenge to the training of decentralized shared models. Some mainstream resampling approaches are widely employed in credit risk prediction because of their good performance in solving the imbalance problem. However, for decentralized training scenarios that require consideration of data security, cost-sensitive learning methods outperform resampling methods in solving the class imbalance problem. Therefore, financial organizations need to focus on developing algorithm-level imbalance handling strategies to enhance the performance of decentralized credit risk prediction methods.

6. Conclusions and future directions

This present work is aimed at addressing the challenges of data privacy protection and unbalanced data distribution in the decentralized modeling of credit risk prediction. For this purpose, this paper develops a FedCSL-CM method based on the HFL paradigm. The proposed method integrates three key components: a CSL-based local model parameter updating method, a CM-based adaptive local model weighting strategy, and an HE-based secure model aggregation mechanism. The performance, business profitability, and data protection capabilities of FedCSL-CM are evaluated and analyzed empirically from four performance measures, two economic measures, and a security perspective. The results demonstrate that FedCSL-CM outperforms local learning approaches, benchmark federated learning approaches, and other unbalanced learning approaches.

In addition, an ablation analysis is conducted. The results indicate that, in decentralized credit risk prediction scenarios, the proposed CSL-based local updating mechanism adapts better to unbalanced data distributions, while the CM-based adaptive weighting strategy evaluates each local model's learning performance effectively and assigns weights rationally. Robustness test results show that increasing the number of training samples improves the predictive performance of the credit risk prediction model significantly, and the proposed method maintains robustness across different local sample sizes. The training

Table A.1
Detailed parameter settings for comparison methods.

Methods	Parameters	Settings
Local learning methods		
LR	max_iter	50
SVM	kernel	"linear"
XGB	n_estimators	30
	max_depth	3
	learning_rate	0.2
LightGBM	n_estimators	30
	max_depth	3
MLP	consistent with the proposed method	
Federated learning methods		
FedAvg	average aggregation	
FedProx	proximal term coefficient	0.001
SCAFFOLD	initial correction parameter	1/ (No. of clients*5)
FedNova	normalized aggregation	
FedKD	distillation temperature parameter	1
	relative weight parameter	0.1

time comparison analysis reveals that FedCSL-CM requires more training time than benchmark federated learning methods, due to its more complex training modules. However, this additional computational cost results in enhanced credit risk prediction performance and stronger data security, making it a valuable tool for risk management professionals to support credit decision-making. We further examined the impact of several key parameters on the performance of the proposed method. The results suggest that the parameters γ and α should not be set too large, to avoid overemphasizing hard-to-classify or minority samples at the expense of overall predictive performance. The λ parameter should be configured under the principle that distillation knowledge does not override local knowledge, ensuring the model remains sensitive to local data distributions. The economic analysis demonstrates that, in multi-organization financial applications, the proposed FedCSL-CM enhances the actual profitability of financial organizations and shows good potential for practical implementation and wide-scale adoption.

There are still some limitations to this paper due to the experimental conditions. (1) We assume that clients and the server will still find ways to obtain data privacy from other clients. However, in a realistic decentralized training scenario, there will be worse cases, such as a participant corrupting the overall decentralized modeling by maliciously uploading false model parameters, resulting in inability to train a useful shared model. Therefore, our assumptions may not fully reflect reality, and there are still some potential risks. (2) We simulate the federated scenario by dividing the experimental dataset equally among participants, which makes the size and distribution of the local data held by each participant roughly the same. In reality, the data samples collected by financial organizations from different regions have significant differences, meaning that our dataset partitioning approach deviates from a truly non-independent and identically distributed (non-IID) setting. Moreover, due to the difficulty of obtaining real-world cross-domain datasets, we did not conduct empirical analyses of the proposed approach across heterogeneous domains. Therefore, the performance of the proposed approach under genuinely non-IID and cross-domain scenarios remains to be further validated.

In future work, we plan to design more secure decentralized learning frameworks to better protect the privacy and security of each participant's private data. Moreover, we will focus on developing new federated learning algorithms that are better adapted to credit risk prediction tasks and more intricately heterogeneous credit data, and evaluating the effectiveness of our approach on real non-IID and cross-domain datasets. Finally, future research can calibrate the key parameters of the proposed method using actual economic costs, such as default losses or opportunity costs, to better align the optimization objectives with real business impacts.

CRediT authorship contribution statement

Jin Xiao: Writing – review & editing, Funding acquisition, Conceptualization. **Zhongyi Wang:** Writing – original draft, Validation, Methodology, Formal analysis, Data curation. **Kuangnan Fang:** Writing – review & editing. **Xiaoyi Jiang:** Methodology. **Dunhu Liu:** Formal analysis. **Shouyang Wang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Parameter setting

Table A.1 lists the specific parameter settings for all comparison methods, including the local learning methods LR, SVM, XGB, LightGBM, and MLP, and the benchmark federated learning methods FedAvg, FedProx, SCAFFOLD, FedNova, and FedKD.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2026.04.006.

Data availability

The data and code are available at https://github.com/wuaoala/FedCSL-CM.

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